

Hello, this is a short presentation on a facial emotion recognition project that I worked on as part of my learning in digital image processing. In this video, I will briefly explain the approach and then demonstrate the working system.

Slide 01:

In this project, I worked on facial emotion recognition, which means identifying human emotions from facial images.

This is a real-world image processing problem because emotions can be subtle, facial images can be low quality, and expressions vary from person to person.

Slide 02:

To solve this problem, I used a Convolutional Neural Network, also known as a CNN.

CNNs are specially designed for image data and can automatically learn important visual features without manual extraction.

A CNN works in three main stages: first, it detects important features from small parts of the image; second, it reduces unnecessary information; and finally, it uses the learned features to classify the emotion.

Slide 03:

The model was trained using the FER2013 dataset, which contains grayscale facial images of size forty-eight by forty-eight pixels.

The dataset includes seven emotion classes and represents real-world conditions with noise and variability.

Due to the complexity of this dataset, an accuracy of around fifty to sixty percent is generally considered acceptable.

Slide 04:

In this project, the images were first preprocessed and augmented, then a CNN model was trained on the dataset.

Finally, the trained model was used to predict emotions on new images.

Through this project, I learned how CNNs work on image data and why real-world emotion recognition is a challenging task.

Code Demo:

I began by importing the required libraries and uploading the FER-2013 dataset. After extracting it, the data was already organized into training and testing folders, where each folder represented a specific facial emotion.

I then defined the dataset paths and preprocessing parameters. All images were resized to forty-eight by forty-eight pixels and normalized. To improve generalization, I applied data augmentation such as rotation, shifting, and horizontal flipping.

Next, I loaded the images using Keras data generators, where emotion labels were automatically inferred from the folder names. I also visualized a few samples to verify that the data was loaded correctly.

I then designed and trained a convolutional neural network from scratch using multiple convolutional and pooling layers, along with batch normalization and dropout to reduce overfitting. The model was trained using the Adam optimizer with early stopping and learning rate reduction, achieving approximately sixty-one percent accuracy on the test dataset.

Now, I will upload a new image and run the trained model on it. As we can see, the model successfully predicted the facial emotion. Finally, I saved the trained model so it can be reused later for inference without retraining.